Exact verification of the dimension-four second pinching inequality

Jinxi Zhang
University of Cambridge

Computational supplement to A proof of the second pinching theorem in dimension four. This notebook explains the finite calculation in Appendix A and calls the accompanying implementation. Its last section executes the complete verification and records the resulting counts.

1. Notation and exact arithmetic

An admissible pair consists of a symmetric trace-free 4 by 4 matrix A and a symmetric cubic tensor tau, both of ordered norm one, with vanishing ordinary and A-weighted cubic traces. In a principal frame A = diag(mu_1,...,mu_4). These are Definition 2.2 and equation (2.11) of the paper. No restriction to distinct or nonzero principal curvatures is imposed.

$$\mathrm{Tr}[A] = 0$$

$$\sum_{i=1}^{4} \mathrm{tau}_{i,i,k} = 0$$

$$\sum_{i=1}^{4} \mathrm{mu}_i \, \mathrm{tau}_{i,i,k} = 0$$

Tensor inner products sum over all ordered indices. A compressed symmetric coordinate with multiplicities n_1,...,n_4 has weight r!/(n_1!...n_4!) for tensors of order r. The implementation retains these weights. Acceptance uses integers and rational numbers; elapsed time and software-version strings are separate output metadata.

Evaluate the input cells in order in a fresh kernel. The package and its data are read relative to this notebook. The named internal routines below are executed by VerifyChern4; their formulas are not reimplemented in this notebook.

In[ ]:=
```
packageDirectory = NotebookDirectory[];
Get[FileNameJoin[{packageDirectory, "Chern4Verifier.wl"}]];
```

2. Spectral and cubic parametrization

Input: (r,y) in the closed unit square and a nonzero vector x of twelve real amplitudes. Output: the unnormalized eigenvalue vector ell, a symmetric cubic T, and the positive normalizations L and N. Routine makeChart constructs these objects. See the spectral and cubic parametrizations (A.2)-(A.5) in Appendix A.

$$\mathrm{ell} = \{1 + r + r\,y,\ 1 - r - r\,y,\ -1 + r - r\,y,\ -1 - r + r\,y\}$$

$$L = \text{ell.ell} = 4\left(1 + r^2 + r^2 y^2\right)$$

The Hadamard map is an isometry from three-dimensional Euclidean space to the trace-free diagonal matrices. Eigenvalue permutations and reversal give all 48 signed permutations of its three coordinates. Arrange their absolute values in decreasing order and divide by the largest. This gives (1,r,ry). When the second coordinate vanishes, take r = y = 0. Routine checkSpectralGroup verifies the finite group identity.

$$C[r, y] = \begin{pmatrix} -1 + r\,y & -r + r\,y \\ 1 + r & 0 \\ 0 & 1 + r \\ -r - r\,y & -1 - r\,y \end{pmatrix}$$

For each k, the vector (T_iik)_i is C(r,y) applied to (x_(2k+3),x_(2k+4)). The four distinct-index entries (T_234,T_134,T_124,T_123) are (x_1,x_2,x_3,x_4). Symmetry gives the other ordered entries. The columns of C lie in the common orthogonal complement of 1 and ell; their minor in rows 2 and 3 is (1+r)^2. They span that complement throughout the square.

$$N = \sum_{i=1}^{4} \sum_{j=1}^{4} \sum_{k=1}^{4} T_{i,j,k}^2$$

$$A = \frac{\text{DiagonalMatrix}[\text{ell}]}{\sqrt{L}}$$

$$\text{tau} = \frac{T}{\sqrt{N}}$$

The repeated-index groups and the distinct-index entries have disjoint ordered supports. The rank statement therefore gives N > 0 when x is nonzero. Also L >= 4. Normalization covers every admissible pair up to the frame and normal-reversal symmetries of the endpoint functional, including repeated and zero eigenvalues.

3. Fourth-tensor contractions and Gram operators

Input: the unnormalized ell and T. Output: the ordinary-trace-free dual Gram matrix and the matrices for the two contractions. Routine rawGram constructs them directly from 35 symmetric fourth-tensor coordinates. Routine checkGramFormulas compares them with the closed matrix formulas of the paper. See the contraction maps (3.1), trace projection (3.7), Gram operators (4.1)-(4.2) and identities (4.4) in the paper.

There are ten symmetric matrix coordinates, nine trace-free matrix coordinates and 35 symmetric fourth-tensor coordinates. Let D_4 be the diagonal matrix of ordered fourth-tensor multiplicities and T_0 the ordinary-trace matrix. The dual Gram matrix of the trace-free orthogonal projection is

$$C_4 = (D_4)^{-1} - (D_4)^{-1} \cdot (T_0)^{T} \cdot \left(T_0 \cdot (D_4)^{-1} \cdot (T_0)^{T}\right)^{-1} \cdot T_0 \cdot (D_4)^{-1}$$

The code checks symmetry, vanishing ordinary trace and C_4 D_4 C_4 = C_4. Applying the contraction matrices to this dual Gram matrix gives the matrix, mixed and vector terms in the joint estimate. The nine-dimensional matrix basis carries its own ordered metric; its inverse enters the contractions in rawPolynomial.

The closed expressions are not used to construct rawGram. They enter only the comparison performed by checkGramFormulas. Both constructions use the same ordered-index convention as the paper. Their intermediate matrices include the positive powers of L and N appropriate to the unnormalized input; equality of the final polynomial is checked after those scalings are removed.

4. Reconstruction of the endpoint polynomial

Input: the symbolic chart and the trial choices in equation (5.2). Output: an exact polynomial P in r,y,x_1,...,x_12. Routine closedPolynomial uses the expanded formula for W_(1/2). Routine rawPolynomial instead assembles the projected fourth-tensor norm and the three zero-average corrections. See the expanded formula (A.1) and polynomial identity (A.6) in Appendix A.

$$P = \frac{5625}{32}\, L^5\, N^2 \left(-W_{\frac{1}{2}}(A, \mathrm{tau}) - \frac{1}{10\,000} \right)$$

Here A and tau are the normalized chart tensors from Section 2. The factor 5625 L^5 N^2/32 is positive for every nonzero amplitude vector. Thus positivity of P implies W_(1/2) <= -1/10000.

P is quartic in the twelve amplitudes and has bidegree at most (14,14) in (r,y). The data contain 196 nonzero quartic channels and 19020 nonzero bivariate coefficient terms. comparePolynomial checks the complete coefficient support: all 1365 quartic types are accounted for, including the 1169 types asserted to vanish. A second check compares the two reconstructed polynomials exactly.

The stored denominator channel is (5625/32)L^5 = 180000(1+r^2+r^2 y^2)^5. It is compared as a polynomial, separately from the numerator channels. No evaluation at sampled spectra enters coefficient acceptance.

In closedPolynomial the matrix ff represents L^2 N B. The auxiliary Python constructor retained in the development archive uses L^3 N B. These are different intermediate arrays. The final scaled P is the object compared in all reconstructions.

5. The rectangular cover and Bernstein expansion

Input: P, its denominator channel and the 24 binary leaf paths in certificate.json. Output: integer-scaled Bernstein arrays on every closed leaf rectangle. Routine checkCover verifies the full binary prefix tree. Routines integerTransforms and affineTransform provide the two conversion methods used by replay. See Lemma A.1 and the Bernstein expansions (A.8) in Appendix A.

At even tree depth a bit bisects the r interval; at odd depth it bisects the y interval. The two children retain the lower and upper closed halves. Every non-leaf prefix has both children, and no listed leaf contains another. The leaf rectangles cover the entire closed square, including its boundary. The literal paths and their order are specified in DATA_SPEC.md.

$$b_i = \sum_{k=0}^{i} \frac{a_k \binom{i}{k}}{\binom{n}{k}}$$

This is the coefficient of the ith Bernstein basis polynomial when the power coefficients are a_k and the degree is n. The calculation uses n = 14 in each variable. Midpoint restriction starts with the factor (14!)^2 and multiplies by 2^14 at each bisection. The direct affine conversion uses (14!)^2 q_r^14 q_y^14, where q_r and q_y are positive endpoint-denominator clearing factors.

For a rational interval [a,b], take a positive integer q clearing both endpoint denominators and set alpha = qa, beta = q(b-a). The integer affine matrix is

$$T_{i,k} = q^{n-k} \sum_{s=0}^{\min(i,k)} \binom{k}{s}\, \mathrm{alpha}^{k-s}\, \mathrm{beta}^{s} \binom{i}{s}\, s!\,(n-s)!$$

It gives n! q^n times the local Bernstein coefficients. In this combinatorial formula a zero base to exponent zero is assigned the value one.

The polynomial and denominator channels always share the same positive integer scale. The 15 by 15 controls on each of the 24 rectangles give 5400 control quartics. Bernstein basis functions are nonnegative and sum to one, so positivity of every control quartic implies positivity on each complete closed rectangle.

Midpoint restriction and direct affine substitution each compute every Bernstein control on the same 24-rectangle cover. Together with the polynomial identities and positive normalization established above, either complete set of Gram identities and positivity checks proves the required bound. The second conversion provides a consistency check on the Bernstein computation. The two methods share the input data and the Gram and integer-positivity routines; their counts are reported separately.

6. Gram identities and integer positive definiteness

Input: each control quartic, its positive denominator coefficient and the supplied Gram correction and congruence witness. Output: the full Gram coefficient identity and positivity of all eight blocks. Routine bookkeeping fixes the ordered monomial positions; verifyLeaf checks all controls. See the Gram reconstruction (A.10), its polynomial identity (A.11) and the integer criterion (A.9) in Appendix A.

The vector z(x) contains the 78 quadratic monomials in twelve amplitudes. Frame-sign characters partition it into blocks of sizes 16,9,9,9,9,9,9,8. The 442 upper-triangular positions carry weight one on the diagonal and two off the diagonal. For a quartic label gamma, c_gamma is the sum of these weights over the positions with that label.

At a fixed control, let p_gamma be the integer-scaled coefficients of P and let d > 0 be the denominator-channel coefficient with the same scale. The Gram matrix G represents (sum_gamma p_gamma x^gamma)/d. Since d > 0, its positivity implies positivity of the polynomial control.

$$G_e = \frac{p_{\mathrm{gamma}(e)}}{c_{\mathrm{gamma}(e)}\ d} + \frac{nu_e}{196\ n_{\mathrm{star}}}$$

The first term is the canonical Gram representation. The second is the bilinear interpolation of four stored zero-polynomial corrections. Their weighted coefficient sums vanish. The common null denominator is n_star = 12884901888. With d > 0, clearing the positive factor 196 n_star d gives the integer matrix Q. DATA_SPEC.md gives the complete interpolation and index conventions.

$$Q_e = \frac{(196\ n_{\mathrm{star}})\ p_{\mathrm{gamma}(e)}}{c_{\mathrm{gamma}(e)}} + d\ nu_e$$

At every control, the code reconstructs all Gram coefficients and checks the denominator. For each symmetric block Q, the supplied integer matrix V is tested by the strict inequalities below. They imply V Q V^T is positive definite, force V to be nonsingular and hence imply Q is positive definite.

$$\left(V.Q.V^{T}\right)_{i,i} > \sum_{j=1}^{i-1} \left|\left(V.Q.V^{T}\right)_{i,j}\right| + \sum_{j=i+1}^{s} \left|\left(V.Q.V^{T}\right)_{i,j}\right|$$

For a nonzero real vector w, the inequality 2|w_i w_j| <= w_i^2+w_j^2 bounds the off-diagonal terms. The strict row inequalities then give w^T V Q V^T w > 0. This is the integer positivity argument used in Appendix A.

Positive definiteness is checked at all 5400 controls, not only at four corners. Since z(x) is nonzero for x != 0, the control quartics are positive. Combined with Section 5 and the positive normalization in Section 4, this proves the required bound. Each replay checks 43200 blocks and 421200 strict row inequalities.

7. Complete execution

The following call starts a new run. It executes both polynomial reconstructions, all data and cover checks, and both complete Bernstein replays. Verbose -> False suppresses progress messages only. Full results are written by the implementation under runs/<run identifier>/; the static data directory is not modified.

In[ ]:=

```
Clear[verification];
verification = Chern4Verifier`VerifyChern4[packageDirectory, "Verbose" → False];
If[! AssociationQ[verification] || ! TrueQ[
    Lookup[verification, "EndpointCertified", False]], Print[verification];
 Abort[]];
```

The counts below are taken from the result of this invocation.

Out[ ]=

Method	Rectangles	Gram identities	Matrix blocks	Row inequalities
Midpoint	24	5400	43200	421200
Affine	24	5400	43200	421200

Out[ ]=

Status	PASS
Endpoint certified	True
Polynomial reconstructions	2
Run identifier	0b23d128-1067-43e2-a204-16348e975df8
Started (UTC)	2026-10-18T00:19:20Z
Finished (UTC)	2026-10-18T00:22:32Z
Wolfram version	15.0.1 for Mac OS X ARM (64-bit) (July 2, 2026)

The command-line entry is wolframscript -file mathematica/run_verification.wls from the supplement root. It performs the same complete verification without a front end. The finite assertions are external inputs to the Lean theorem; a successful Wolfram run does not supply a Lean proof term for them.

Out[ ]=

```
<|Status -> PASS, EndpointCertified -> True,
>   TotalIntegerMatrixChecks -> 86400|>
```